

the product σ -algebra $\mathcal{B}^{(K)}$, where $\nu(A) = \frac{1}{K}|A|$ is the normalized counting measure defined on any subset $A \subseteq \{1, \dots, K\}$, and a $\mu^{(K)}$ -preserving transformation $T^{(K)}$ by

$$T^{(K)}(x, i) = \begin{cases} (x, i+1) & \text{if } 1 \leq i < K, \\ (Tx, 1) & \text{if } i = K \end{cases}$$

for all $x \in X$. Show that $T^{(K)}$ is ergodic with respect to $\mu^{(K)}$ if and only if T is ergodic with respect to μ , and that $T^{(K)}$ is not totally ergodic if $K > 1$.

2.6 Pointwise Ergodic Theorem

The conventional proof of the pointwise ergodic theorem involves two other important results, the maximal inequality and the maximal ergodic theorem. Roughly speaking, the maximal ergodic theorem may be used to show that the set of functions in L^1_μ for which the pointwise ergodic theorem holds is *closed* as a subset of L^1_μ ; one then has to find a *dense* subset of L^1_μ for which the pointwise ergodic theorem holds. Examples 2.23 and 2.25 give another motivation for the maximal ergodic theorem.

Since the pointwise ergodic theorem involves evaluating a function along the orbit of individual points, it is most naturally phrased in terms of genuine functions (that is, elements of $\mathcal{L}^1_\mu$; see Sect. A.3 for the notation). We will normally apply it to a function in L^1_μ , where the meaning is that for any representative in $\mathcal{L}^1_\mu$ of the equivalence class in L^1_μ we have convergence almost everywhere.

2.6.1 The Maximal Ergodic Theorem

In order to see where the next result comes from, it is useful to ask how likely is it that the orbit of a point spends unexpectedly much time in a given small set (the ergodic theorem says that the orbit of a point spends a predictable amount of time in a given set).

Example 2.23. Let $(X, \mathcal{B}_X, \mu, T)$ be a measure-preserving system, and fix a small measurable set $B \in \mathcal{B}_X$ with $\mu(B) = \varepsilon > 0$. Consider the ergodic average

$$A_N^{\chi_B} = \frac{1}{N} \sum_{n=0}^{N-1} \chi_B \circ T^n.$$

Since T preserves μ , $\int_X \chi_B \circ T^n d\mu = \mu(B)$ for any $n \geq 0$, so

$$\int_X A_N^{\chi_B} d\mu = \int_X \chi_B d\mu = \mu(B) = \varepsilon.$$

Now ask how likely is it that the orbit of a point x spends more than $\sqrt{\varepsilon} > \varepsilon$ of the time between 0 and $N - 1$ in the set B . Notice that

$$\sqrt{\varepsilon} \mu(\{x \mid A_N^{\chi_B}(x) > \sqrt{\varepsilon}\}) \leq \int_X A_N^{\chi_B} d\mu = \varepsilon,$$

since

$$\sqrt{\varepsilon} \chi_{\{y \mid A_N^{\chi_B}(y) > \sqrt{\varepsilon}\}}(x) \leq A_N^{\chi_B}(x)$$

for all $x \in X$. Thus on the fixed time scale $[0, N - 1]$ the measure of the set B_ε^N of points that spend in proportion at least $\sqrt{\varepsilon}$ of the time between 0 and $N - 1$ in the set B is no larger than $\sqrt{\varepsilon}$.

We would like to be able to say that one can find a set B_ε independent of N with similar properties for all N ; as discussed below, this is a consequence of the maximal ergodic theorem⁽²⁰⁾.

Theorem 2.24 (Maximal Ergodic Theorem). *Consider the measure-preserving system $(X, \mathcal{B}, \mu, T)$ on a probability space and g a real-valued function in $\mathcal{L}_\mu^1$. Define*

$$E_\alpha = \left\{ x \in X \mid \sup_{n \geq 1} \frac{1}{n} \sum_{i=0}^{n-1} g(T^i x) > \alpha \right\}$$

for any $\alpha \in \mathbb{R}$. Then

$$\alpha \mu(E_\alpha) \leq \int_{E_\alpha} g d\mu \leq \|g\|_1.$$

Moreover, $\alpha \mu(E_\alpha \cap A) \leq \int_{E_\alpha \cap A} g d\mu$ whenever $T^{-1}A = A$.

Example 2.25. We continue the discussion from Example 2.23 by noting that if $B \subseteq X$ has $\mu(B) = \varepsilon > 0$ and $g = \chi_B$ is its characteristic function, then by applying the maximal ergodic theorem (Theorem 2.24) with $\alpha = \sqrt{\varepsilon}$ we get the following statement: There exists a set $B' \subseteq X$ with $\mu(B') \leq \sqrt{\varepsilon}$ such that for all $N \geq 1$ and all $x \in X \setminus B'$ the orbit of the point x spends at most $\sqrt{\varepsilon}$ in proportion of the times between 0 and $N - 1$ in the set B . Thus we have found a set as in Example 2.23, but independently of N .

2.6.2 Maximal Ergodic Theorem via Maximal Inequality

Notice that the operator U_T associated to a measure-preserving transformation T is a *positive* linear operator on each L_μ^p space (positive means

that $f \geq 0$ implies $U_T f \geq 0$). A traditional proof of Theorem 2.24 starts with a maximal inequality for positive operators.

Proposition 2.26 (Maximal Inequality). *Let $U : L_\mu^1 \rightarrow L_\mu^1$ be a positive linear operator with $\|U\| \leq 1$. For $f \in L_\mu^1$ a real-valued function, define inductively the functions*

$$\begin{aligned} f_0 &= 0 \\ f_1 &= f \\ f_2 &= f + Uf \\ &\vdots \\ f_n &= f + Uf + \cdots + U^{n-1}f \end{aligned}$$

for $n \geq 1$, and $F_N = \max\{f_n \mid 0 \leq n \leq N\}$ (all these functions are defined pointwise). Then

$$\int_{\{x \mid F_N(x) > 0\}} f \, d\mu \geq 0$$

for all $N \geq 1$.

PROOF. For each N , it is clear that $F_N \in L_\mu^1$. Since U is positive and linear, and since

$$F_N \geq f_n$$

for $0 \leq n \leq N$, we have

$$UF_N + f \geq Uf_n + f = f_{n+1}.$$

Hence

$$UF_N + f \geq \max_{1 \leq n \leq N} f_n.$$

For $x \in P = \{x \mid F_N(x) > 0\}$ we have

$$F_N(x) = \max_{0 \leq n \leq N} f_n(x) = \max_{1 \leq n \leq N} f_n(x)$$

since $f_0 = 0$. Therefore,

$$UF_N(x) + f(x) \geq F_N(x)$$

for $x \in P$, and so

$$f(x) \geq F_N(x) - UF_N(x) \tag{2.13}$$

for $x \in P$. Now $F_N(x) \geq 0$ for all x , so $UF_N(x) \geq 0$ for all x . Hence the inequality (2.13) implies that

$$\begin{aligned}
\int_P f \, d\mu &\geq \int_P F_N \, d\mu - \int_P U F_N \, d\mu \\
&= \int_X F_N \, d\mu - \int_P U F_N \, d\mu \quad (\text{since } F_N(x) = 0 \text{ for } x \notin P) \\
&\geq \int_X F_N \, d\mu - \int_X U F_N \, d\mu \\
&= \|F_N\|_1 - \|U F_N\|_1 \geq 0,
\end{aligned}$$

since $\|U\| \leq 1$. □

FIRST PROOF OF THEOREM 2.24. Let $f = (g - \alpha)$ and $Uf = f \circ T$ for $f \in \mathcal{L}_\mu^1$ so that, in the notation of Proposition 2.26,

$$E_\alpha = \bigcup_{N=0}^{\infty} \{x \mid F_N(x) > 0\}.$$

It follows that $\int_{E_\alpha} f \, d\mu \geq 0$ and therefore $\int_{E_\alpha} g \, d\mu \geq \alpha \mu(E_\alpha)$. For the last statement, apply the same argument to $f = (g - \alpha)$ on the measure-preserving system $(A, \mathcal{B}|_A, \frac{1}{\mu(A)}\mu|_A, T|_A)$. □

2.6.3 Maximal Ergodic Theorem via a Covering Lemma

In this subsection we use covering properties of intervals in $\mathbb{Z}$ to establish a version of the maximal ergodic theorem (Theorem 2.24). This demonstrates very clearly the strong link between the Lebesgue density theorem (Theorem A.24), whose proof involves the Hardy–Littlewood maximal inequality, and the pointwise ergodic theorem, whose proof involves the maximal ergodic theorem*. The material in this section illustrates some of the ideas used in the more extensive results of Bourgain [41]; a little of the history will be given in the note (83) on p. 275.

We will obtain a formally weaker version of Theorem 2.24, by showing that

$$\alpha \mu(E_\alpha) \leq 3 \|g\|_1 \tag{2.14}$$

in the notation of Theorem 2.24. This is sufficient for all our purposes. For future applications, we state the covering lemma⁽²¹⁾ needed in a more general setting.

Lemma 2.27 (Finite Vitali covering lemma). *Let $B_{r_1}(a_1), \dots, B_{r_K}(a_K)$ be any collection of balls in a metric space. Then there exists a subcollec-*

* Additionally, this approach starts to reveal more about what properties of the acting group might be useful for obtaining more general ergodic theorems, and gives a method capable of generalization to ergodic averaging along other sets of integers.

tion $B_{r_{j(1)}}(a_{j(1)}), \dots, B_{r_{j(k)}}(a_{j(k)})$ of those balls which are disjoint and satisfy

$$B_{r_1}(a_1) \cup \dots \cup B_{r_K}(a_K) \subseteq B_{3r_{j(1)}}(a_{j(1)}) \cup \dots \cup B_{3r_{j(k)}}(a_{j(k)}),$$

where in the right-hand side we have tripled the radii of the balls in the subcollection.

PROOF. By reordering the balls if necessary, we may assume that

$$r_1 \geq r_2 \geq \dots \geq r_K.$$

Let $j(1) = 1$. We choose the remaining disjoint balls by induction as follows. Assume that we have chosen $j(1), \dots, j(n)$ from the indices $\{1, \dots, \ell\}$, discarding those not chosen. If $B_{r_{\ell+1}}(a_{\ell+1})$ is disjoint from

$$B_{r_{j(1)}}(a_{j(1)}) \cup \dots \cup B_{r_{j(n)}}(a_{j(n)})$$

we choose $j(n+1) = \ell+1$, and if not we discard $\ell+1$, and proceed with studying $\ell+2$, stopping if $\ell+1 = K$. Suppose that $B_{r_{j(1)}}(a_{j(1)}), \dots, B_{r_{j(k)}}(a_{j(k)})$ are the balls chosen from all the balls considered, and let

$$V = B_{3r_{j(1)}}(a_{j(1)}) \cup \dots \cup B_{3r_{j(k)}}(a_{j(k)}).$$

If $i \in \{j(1), \dots, j(k)\}$ then $B_{r_i}(a_i) \subseteq B_{3r_i}(a_i) \subseteq V$ by construction. If not, then by the construction there is some $n \in \{1, \dots, i-1\} \cap \{j(1), \dots, j(k)\}$ that was selected, such that

$$B_{r_i}(a_i) \cap B_{r_n}(a_n) \neq \emptyset,$$

and $r_n \geq r_i$ by the ordering of the indices. By the triangle inequality we therefore have

$$B_{r_i}(a_i) \subseteq B_{3r_n}(a_n) \subseteq V$$

as required. $\square$

In the integers, the Vitali covering lemma may be formulated as follows (see Exercise 2.6.2).

Corollary 2.28. *For any collection of intervals*

$$I_1 = [a_1, a_1 + \ell(1) - 1], \dots, I_K = [a_K, a_K + \ell(K) - 1]$$

in $\mathbb{Z}$ there is a disjoint subcollection $I_{j(1)}, \dots, I_{j(k)}$ such that

$$I_1 \cup \dots \cup I_K \subseteq \bigcup_{m=1}^k [a_{j(m)} - \ell(j(m)), a_{j(m)} + 2\ell(j(m)) - 1].$$

PROOF OF THE INEQUALITY (2.14). Let $(X, \mathcal{B}, \mu, T)$ be a measure-preserving system, with $g \in \mathcal{L}_\mu^1$, and fix $\alpha > 0$. Define

$$g^*(x) = \sup_{n \geq 1} \frac{1}{n} \sum_{i=0}^{n-1} g(T^i(x))$$

and $E_\alpha = \{x \in X \mid g^*(x) > \alpha\}$ as before. We will deduce the inequality (2.14) from a similar estimate for the function

$$\phi(j) = \begin{cases} g(T^j x) & \text{for } j = 0, \dots, J; \\ 0 & \text{for } j < 0 \text{ or } j > J \end{cases} \quad (2.15)$$

for a fixed $x \in X$ and $J \geq 1$.

Lemma 2.29. *For any $\phi \in \ell^1(\mathbb{Z})$ and $\alpha > 0$, define*

$$\phi^*(a) = \sup_{n \geq 1} \frac{1}{n} \sum_{i=0}^{n-1} \phi(a+i),$$

and

$$E_\alpha^\phi = \{a \in \mathbb{Z} \mid \phi^*(a) > \alpha\}.$$

Then $\alpha |E_\alpha^\phi| \leq 3 \|\phi\|_1$.

PROOF OF LEMMA 2.29. Let $a_1, \dots, a_K$ be different elements of E_α^ϕ , and let $\ell(j)$ for $j = 1, \dots, K$ be chosen so that

$$\frac{1}{\ell(j)} \sum_{i=0}^{\ell(j)-1} \phi(a_j + i) > \alpha. \quad (2.16)$$

Define the intervals $I_j = [a_j, a_j + \ell(j) - 1]$ for $1 \leq j \leq K$ and use Corollary 2.28 to construct the subcollection $I_{j(1)}, \dots, I_{j(k)}$ as in the corollary. Since the intervals $I_{j(1)}, \dots, I_{j(k)}$ are disjoint, it follows that

$$\sum_{i=1}^k \sum_{m \in I_{j(i)}} \phi(m) \leq \|\phi\|_1, \quad (2.17)$$

where the left-hand side equals

$$\sum_{i=1}^k \ell(j(i)) \frac{1}{\ell(j(i))} \sum_{n=0}^{\ell(j(i))-1} \phi(a_j + n) > \sum_{i=1}^k \ell(j(i)) \alpha \quad (2.18)$$

by the choice in (2.16) of the $\ell(j(i))$. However, since

$$\{a_1, \dots, a_K\} \subseteq \bigcup_{j=1}^k [a_{j(i)} - \ell(j(i)), a_{j(i)} + 2\ell(j(i)) - 1]$$

by Corollary 2.28, we therefore have

$$K \leq 3 \sum_{i=1}^k \ell(j(i)). \quad (2.19)$$

Combining the inequalities (2.19), (2.18), and (2.17) in that order gives

$$\alpha K \leq 3 \sum_{i=1}^k \ell(j(i))\alpha < 3\|\phi\|_1,$$

which proves the lemma. $\square$

Fix now some $M \geq 1$ (the parameter J will later be chosen much larger than M) and define

$$g_M^*(x) = \sup_{1 \leq n \leq M} \frac{1}{n} \sum_{i=0}^{n-1} g(T^i x),$$

and

$$E_{\alpha, M}^g = \{x \in X \mid g_M^*(x) > \alpha\}.$$

Using ϕ as in (2.15) and, suppressing the dependence on x as before, we also define

$$\phi_M^*(a) = \sup_{1 \leq n \leq M} \frac{1}{n} \sum_{i=0}^{n-1} \phi(a+i).$$

As $\phi(a+i) = g(T^{a+i}x)$ if $0 \leq a < J-M$ and $0 \leq i < M$, we have

$$\phi_M^*(a) = g_M^*(T^a x) \quad (2.20)$$

for $0 \leq a < J-M$. Also, for any $x \in X$ and $\alpha > 0$ we have

$$\alpha |\{a \in [0, J-1] \mid \phi_M^*(a) > \alpha\}| \leq 3\|\phi\|_1$$

by Lemma 2.29. Recalling the definition of ϕ and E_α and using (2.20), this may be written in a slightly weaker form as

$$\begin{aligned} \alpha \sum_{a=0}^{J-M-1} \chi_{E_{\alpha, M}^g}(T^a x) &= \alpha \left| \left\{ a \in [0, J-M-1] \mid g_M^*(T^a x) > \alpha \right\} \right| \\ &\leq 3 \sum_{i=0}^J |g(T^i x)|, \end{aligned}$$

which may be integrated over $x \in X$ to obtain

$$(J-M)\alpha\mu\left(E_{\alpha, M}^g\right) \leq 3(J+1)\|g\|_1,$$

where we have used the invariance of μ under T . Dividing by J and letting $J \rightarrow \infty$ gives $\alpha\mu(E_{\alpha,M}^g) \leq 3\|g\|_1$, and finally letting $M \rightarrow \infty$ gives inequality (2.14). $\square$

2.6.4 The Pointwise Ergodic Theorem

We are now ready to give a proof of Birkhoff's pointwise ergodic theorem [33] using the maximal ergodic theorem⁽²²⁾. This precisely describes the relationship sought between the space average of a function and the time average along the orbit of a typical point.

Theorem 2.30 (Birkhoff). *Let $(X, \mathcal{B}, \mu, T)$ be a measure-preserving system. If $f \in \mathcal{L}_\mu^1$, then*

$$\lim_{n \rightarrow \infty} \frac{1}{n} \sum_{j=0}^{n-1} f(T^j x) = f^*(x)$$

converges almost everywhere and in L_μ^1 to a T -invariant function $f^ \in \mathcal{L}_\mu^1$, and*

$$\int f^* d\mu = \int f d\mu.$$

If T is ergodic, then

$$f^*(x) = \int f d\mu$$

almost everywhere.

Example 2.31. ⁽²³⁾ In Example 1.2 we explained that almost every real number has the property that any block of length k of digits base 10 appears with asymptotic frequency $\frac{1}{10^k}$, thus almost every number is *normal* base 10. We now have all the material needed to justify this result: By Corollary 2.20, the map $x \mapsto Kx \pmod{1}$ on the circle for $K \geq 2$ is ergodic, so the pointwise ergodic theorem (Theorem 2.30) may be applied to show that almost every number is normal to each base $K \geq 2$, and so (by taking the union of countably many null sets) almost every number is normal in *every* base $K \geq 2$.

As with the maximal ergodic theorem (Theorem 2.24), we will give two proofs⁽²⁴⁾ of the pointwise ergodic theorem. The first is a traditional one while the second is closer to the approach of Bourgain [41] for example, and is better adapted to generalization both of the acting group and of the sequence along which ergodic averages are formed.

Theorem 2.30 will be formulated differently in Theorem 6.1, and will be used in Theorem 6.2 to construct the ergodic decomposition.

2.6.5 Two Proofs of the Pointwise Ergodic Theorem

FIRST PROOF OF THEOREM 2.30. Recall that $(X, \mathcal{B}, \mu, T)$ is a measure-preserving system, $\mu(X) = 1$, and $f \in \mathcal{L}_\mu^1$. It is sufficient to prove the result for a real-valued function f . Define, for any $x \in X$,

$$f^*(x) = \limsup_{n \rightarrow \infty} \frac{1}{n} \sum_{i=0}^{n-1} f(T^i x),$$

$$f_*(x) = \liminf_{n \rightarrow \infty} \frac{1}{n} \sum_{i=0}^{n-1} f(T^i x).$$

Then

$$\frac{n+1}{n} \left(\frac{1}{n+1} \sum_{i=0}^n f(T^i x) \right) = \frac{1}{n} \sum_{i=0}^{n-1} f(T^i(Tx)) + \frac{1}{n} f(x). \quad (2.21)$$

By taking the limit along a subsequence for which the left-hand side of (2.21) converges to the limsup, this shows that $f^* \leq f^* \circ T$. A limit along a subsequence for which the right-hand side of (2.21) converges to the limsup shows that $f^* \geq f^* \circ T$. A similar argument for f_* shows that

$$f^* \circ T = f^*, \quad f_* \circ T = f_*. \quad (2.22)$$

Now fix rationals $\alpha > \beta$, and write

$$E_\alpha^\beta = \{x \in X \mid f_*(x) < \beta \text{ and } f^*(x) > \alpha\}.$$

By (2.22), $T^{-1}E_\alpha^\beta = E_\alpha^\beta$ and $E_\alpha \supseteq E_\alpha^\beta$ where E_α is the set defined in Theorem 2.24 (with $g = f$). By Theorem 2.24,

$$\int_{E_\alpha^\beta} f \, d\mu \geq \alpha \mu(E_\alpha^\beta). \quad (2.23)$$

After replacing f by $-f$, a similar argument shows that

$$\int_{E_\alpha^\beta} f \, d\mu \leq \beta \mu(E_\alpha^\beta). \quad (2.24)$$

Now

$$\{x \mid f_*(x) < f^*(x)\} = \bigcup_{\substack{\alpha, \beta \in \mathbb{Q}, \\ \alpha > \beta}} E_\alpha^\beta,$$

while the inequalities (2.23) and (2.24) show that $\mu(E_\alpha^\beta) = 0$ for $\alpha > \beta$. It follows that

$$\mu \left(\bigcup_{\substack{\alpha, \beta \in \mathbb{Q}, \\ \alpha > \beta}} E_{\alpha}^{\beta} \right) = 0,$$

so

$$f_*(x) = f^*(x) \text{ a.e.}$$

Thus

$$g_n(x) = \frac{1}{n} \sum_{i=0}^{n-1} f(T^i x) \longrightarrow f^*(x) \text{ a.e.} \quad (2.25)$$

By Corollary 2.22 we also know that

$$g_n \xrightarrow{L_{\mu}^1} f' \in \mathcal{L}_{\mu}^1. \quad (2.26)$$

By Corollary A.12, this implies that there is a subsequence $n_k \rightarrow \infty$ with

$$g_{n_k}(x) \longrightarrow f'(x) \text{ a.e.} \quad (2.27)$$

Putting (2.25), (2.26) and (2.27) together we see that $f^* = f' \in \mathcal{L}_{\mu}^1$ and that the convergence in (2.25) also happens in L_{μ}^1 . Finally we also get

$$\int f \, d\mu = \int g_n \, d\mu = \int f^* \, d\mu.$$

□

A somewhat different approach is to use the maximal ergodic theorem (Theorem 2.24) to control the gap between mean convergence and pointwise convergence almost everywhere.

SECOND PROOF OF THEOREM 2.30. Assume first that $f_0 \in \mathcal{L}^{\infty}$. By the mean ergodic theorem in L^1 (Corollary 2.22) we know that the ergodic averages

$$A_N(f_0) = \frac{1}{N} \sum_{n=0}^{N-1} f_0 \circ T^n \rightarrow F_0$$

converge in L_{μ}^1 as $N \rightarrow \infty$ to some T -invariant function $F_0 \in \mathcal{L}_{\mu}^1$. Given $\varepsilon > 0$ choose some M such that

$$\|F_0 - A_M(f_0)\|_1 < \varepsilon^2.$$

By the maximal ergodic theorem (Theorem 2.24) applied to the function

$$g(x) = F_0(x) - A_M(f_0)$$

we see that

$$\varepsilon \mu \left(\{x \in X \mid \sup_{N \geq 1} |A_N(F_0 - A_M(f_0))| > \varepsilon\} \right) < \varepsilon^2.$$

Clearly $A_N(F_0) = F_0$ since the limit function F_0 is T -invariant, while if M is fixed and $N \rightarrow \infty$ we have (see Exercise 2.6.4)

$$\begin{aligned} A_N(A_M(f_0)) &= \frac{1}{NM} \sum_{n=0}^{N-1} \sum_{m=0}^{M-1} f_0 \circ T^{n+m} \\ &= A_N(f_0) + O_M\left(\frac{\|f_0\|_\infty}{N}\right). \end{aligned} \quad (2.28)$$

Putting these together, we see that

$$\begin{aligned} \mu(\{x \mid \limsup_{N \rightarrow \infty} |F_0 - A_N(f_0)| > \varepsilon\}) &= \mu(\{x \mid \limsup_{N \rightarrow \infty} |F_0 - A_N(A_M(f_0))| > \varepsilon\}) \\ &\leq \mu(\{x \mid \sup_{N \geq 1} |A_N(F_0 - A_M(f_0))| > \varepsilon\}) \\ &< \varepsilon, \end{aligned}$$

which shows that $A_N(f_0) \rightarrow F_0$ almost everywhere.

To prove convergence for any $f \in \mathcal{L}_\mu^1$, fix $\varepsilon > 0$ and choose some $f_0 \in \mathcal{L}^\infty$ with $\|f - f_0\|_1 < \varepsilon^2$. Write $F \in \mathcal{L}_\mu^1$ for the L^1 -limit of $A_N(f)$ and $F_0 \in \mathcal{L}_\mu^1$ for the L^1 -limit of $A_N(f_0)$. Since $\|A_N(f) - A_N(f_0)\|_1 \leq \|f - f_0\|_1$ we deduce that $\|F - F_0\|_1 < \varepsilon^2$. From this we get

$$\begin{aligned} &\mu(\{x \mid \limsup_{N \rightarrow \infty} |F - A_N(f)| > 2\varepsilon\}) \\ &\leq \mu(\{x \mid |F - F_0| + \limsup_{N \rightarrow \infty} |F_0 - A_N(f_0)| + \sup_{N \geq 1} |A_N(f_0 - f)| > 2\varepsilon\}) \\ &\leq \mu(\{x \mid |F - F_0| > \varepsilon\}) + \mu(\{x \mid \sup_{N \geq 1} |A_N(f_0 - f)| > \varepsilon\}) \\ &\leq \varepsilon^{-1} \|F - F_0\|_1 + \varepsilon^{-1} \|f_0 - f\|_1 \leq 2\varepsilon \quad (2.29) \end{aligned}$$

by the maximal ergodic theorem (Theorem 2.24), which shows that $A_N(f)$ converges almost everywhere as $N \rightarrow \infty$. $\square$

Exercises for Sect. 2.6

Exercise 2.6.1. Prove the following version of the ergodic theorem for finite permutations (see the book of Nadkarni [263] where this is used to motivate a different approach to ergodic theorems). Let $X = \{x_1, \dots, x_r\}$ be a finite set, and let $\sigma : X \rightarrow X$ be a permutation of X . The orbit of x_j under σ is the set $\{\sigma^n(x_j)\}_{n \geq 0}$, and σ is called cyclic if there is an orbit of cardinality r .

(1) For a cyclic permutation σ and any function $f : X \rightarrow \mathbb{R}$, prove that